

Lambda Trigger

To Begin the Lab

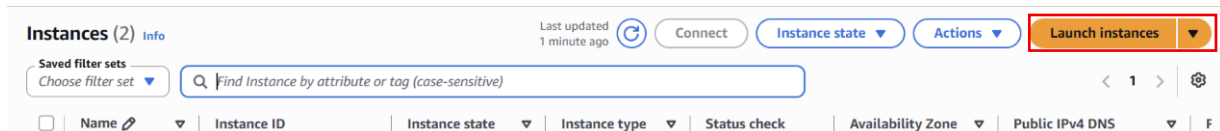
Summary of the Lab

This lab demonstrates how to trigger AWS Lambda functions automatically using **API Gateway** and EventBridge. You create an API Gateway trigger to start an EC2 instance from a web page and configure an EventBridge scheduled rule using a cron expression to stop the instance automatically at specified time intervals, introducing event-driven automation in AWS.

Lab steps

Step 1: Launch a New Instance

- Go to the EC2 Instance
- Click on the **launch Instance**



Step 2: Enter Instance Details

- Provide the Instance Name to identify the instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

Step 3: Choose an Amazon Machine Image (AMI)

- Choose Amazon Linux AWS

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

 Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start




[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-0d351f1b760a30161 (64-bit (x86), uefi-preferred) / ami-0a2f3ecf7b370b274 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Step 4: Choose an Instance Type

- Choose the Instance Type t2.micro

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro
Family: T2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.017 USD per Hour On-Demand RHEL base pricing: 0.0268 USD per Hour
On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Step 5: Create or Select a Key Pair

- Choose the key pair, that who If key pair created before or Create New key pair

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Mumbai

 [Create new key pair](#)

Step 6: Configure Network Settings

- Go to the Network section
- Choose Create security Group

- Select the Allow SSH traffic from

Network [Info](#)

vpc-072c4acd761c3b942

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1186' with the following rules:

☒ Allow SSH traffic from Anywhere
0.0.0.0/0

Helps you connect to your instance

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

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Step 7: Launch the Instance

- Click on the Launch Instance

Firewall (security groups) [Info](#)

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To set up an endpoint, for example when creating a web server

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#)

- EC2 Instance Create successfully
- Copy Instance id for Lambda

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	myweb	i-060560e011f44125a	Running	t2.micro	2/2 checks passed	ap-south-1a

- Go to the instance state, stop the instance

Instances (1/3) [Info](#)

Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	myweb	i-03b89b7e63d3278a4	Running	t3.micro	Initializing	us-east-1a	ec2-3-85-143-18

[Stop instance](#) [Start instance](#) [Reboot instance](#) [Hibernate instance](#)

[Elastic IP](#) [IPv6 IPs](#)

Step 8: Create the IAM Role

- Go the IAM Role in your AWS account
- Create Role
- Select AWS service

- Choose Use case Lambda

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**
 Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
 Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
 Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
 Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
 Create a custom trust policy to enable others to perform actions in this account.

Use case
 Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

☒ **Lambda**
 Allows Lambda functions to call AWS services on your behalf.

☐ **AWSServiceRoleForLambda**
 Allows Lambda to manage AWS resources on your behalf.

- Click on the Next

Use case

☒ **Lambda**
 Allows Lambda functions to call AWS services on your behalf.

☐ **AWSServiceRoleForLambda**
 Allows Lambda to manage AWS resources on your behalf.

Cancel **Next**

Step 9: Add Permission Policies

- Select two policies
- AWSLambdaBasicExecutionRole And AmazonEC2FullAccess
- Click on the Next

Add permissions [Info](#)

Permissions policies (1/2312) [Info](#)

Choose one or more policies to attach to your new role.

Filter by Type

Q AWSLambdaBasicExecutionRole X All types 446 matches

Policy name	Type	Description
☒ AWSLambdaBasicExecutionRole	AWS managed	Provides write permissions to CloudW...

Filter by Type

Q AmazonEC2Full X All types 1 match

Policy name	Type	Description
☒ AmazonEC2FullAccess	AWS managed	Provides full access to Amazon EC2 via th...

► Set permissions boundary - optional

Cancel Previous **Next**

Step 10: Name the Role

- Provide a Role Name to identify the role
- This role Use in Lambda Function, then to the Following

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+=,._-' characters.

- Click on the Create Role

Step 3: Add tags

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

Cancel Previous **Create role**

Step 11: Create a Function

- Go to the Lambda in your AWS account
- Click on the Create Function

Functions (153)

Functions table allows you to query function details. Use Ctrl + K to start a in-console resource search.

Q Search by attributes or search by keyword 1 match

Last fetched 7/1/2026, 11:28:02 AM [Actions](#) **Create function**

Step 12: Configure the Function

- Select Author from scratch
- Provide the function Name **StartEC2Instance** to identify the function
- Choose the Runtime Python

Create function [Info](#)

Choose one of the following options to create your function.

☒ Author from scratch

Start with a simple Hello World example.

☐ Use a blueprint

Build a Lambda application from sample code and configuration presets for common use cases.

☐ Container image

Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

StartEC2instance

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.13

Step 13: Choose an Execution Role

- Custom Execution role Allow
- Choose the before created **LambdaEC2Role**
- Click on the Save

Additional settings

Customize your function architecture, execution role, HTTP(S) endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

General [Info](#)

ARM64 architecture

Use ARM64 instead of x86_64 (default).



Custom execution role

Use custom execution role instead of new role with basic Lambda permissions (default).



Networking [Info](#)

Function URL

Assign HTTP(S) endpoints.



Tenant isolation mode - [new](#)

Guarantee separate execution environments for each tenant invoking the function.



Configure custom execution role

Choose an existing execution role or create a new one.

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

LambdaEC2Role



Create new role

Edit role

[View role details in IAM](#)

Close

Save

- Click on the Create Function

Cannot add or remove after creation

Cancel

Create function

- Function Create Successfully
- Click on the Function

Functions (153)

Functions table allows you to query function details. Use Ctrl + K to start a in-console resource search.

Search by attributes or search by keyword

1 match

Function name = StartEC2instance



Clear filters



Function name



Description



Package type



Runtime



StartEC2instance

-

Zip

Python 3.13

Step 14: Deploy

- Go to function code source
- Implement this code to Start instance

```
import boto3
ec2 = boto3.client('ec2')
INSTANCE_ID = "YOUR_INSTANCE_ID"
def lambda_handler(event, context):
    ec2.start_instances(Deploy
```

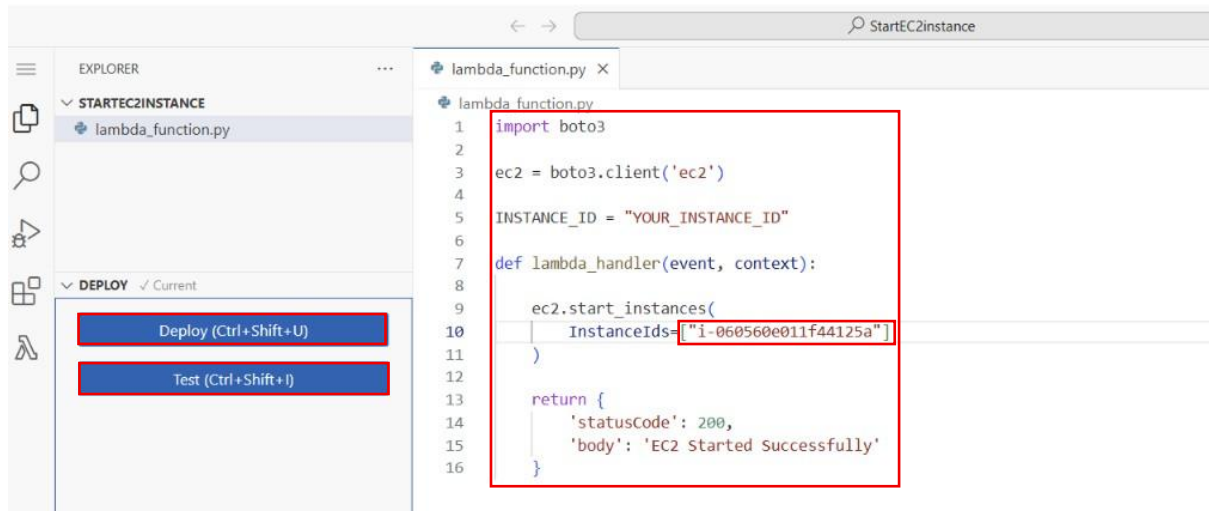
```

InstanceIds=[" "]
)
return {
    'statusCode': 200,
    'body': 'EC2 Started Successfully'
}

```

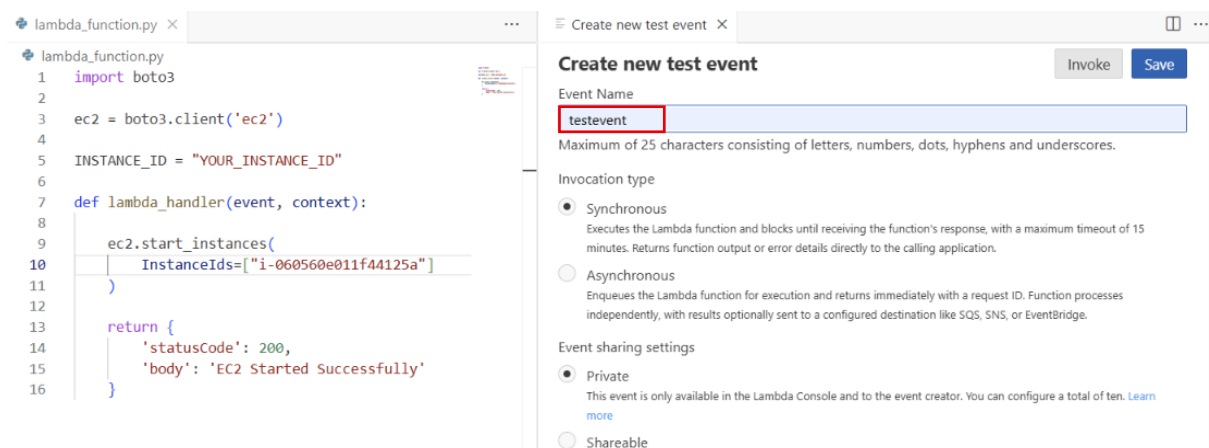
- Paste your instance Id in Code
- Click on the Deploy

Code source [Info](#)

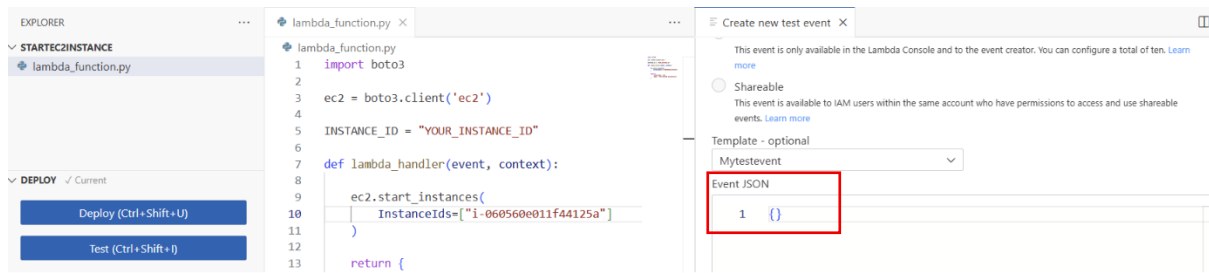


Step 15: Create Test Event

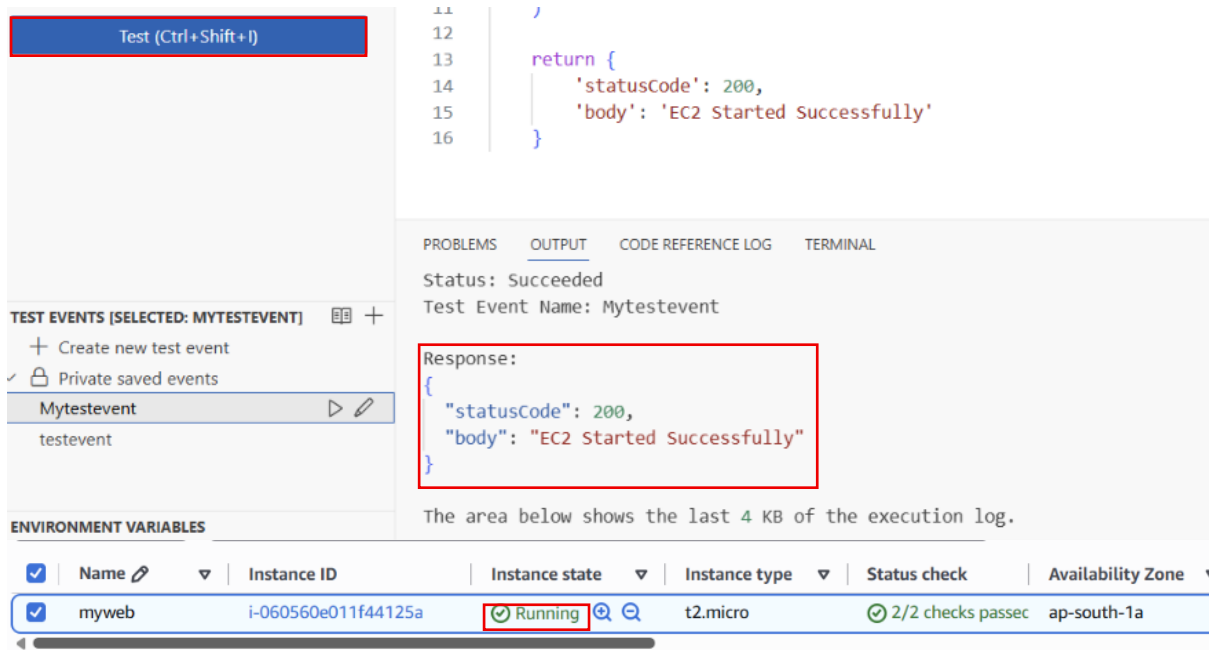
- Click on the Test
- Enter the Event Name



- This is simple Test Event
- Save the event



- Click again on the Test
- Function is Ready to Start instance
- Check the EC2 instance is Start

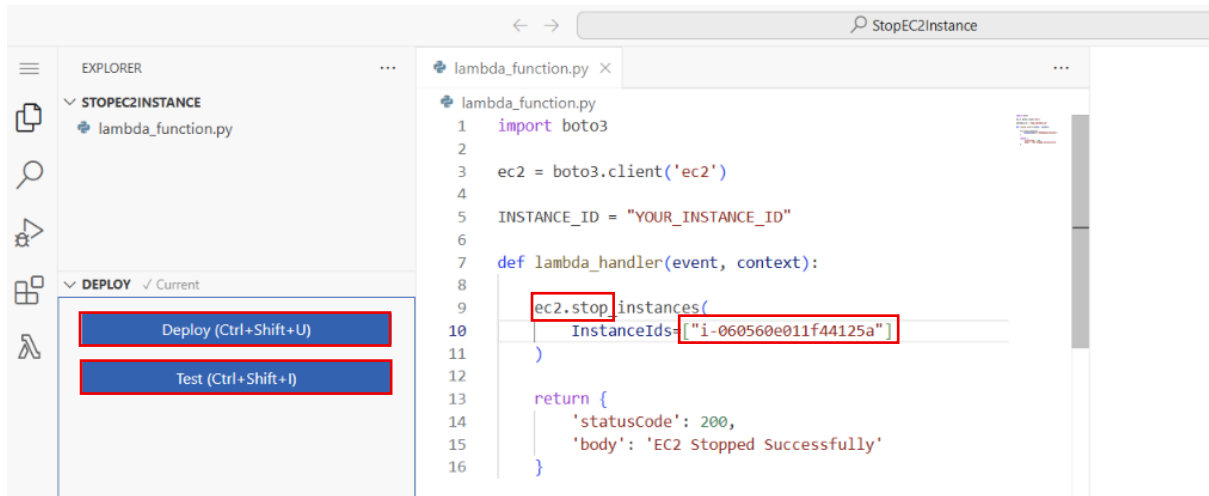


Step 16: Create stop Function

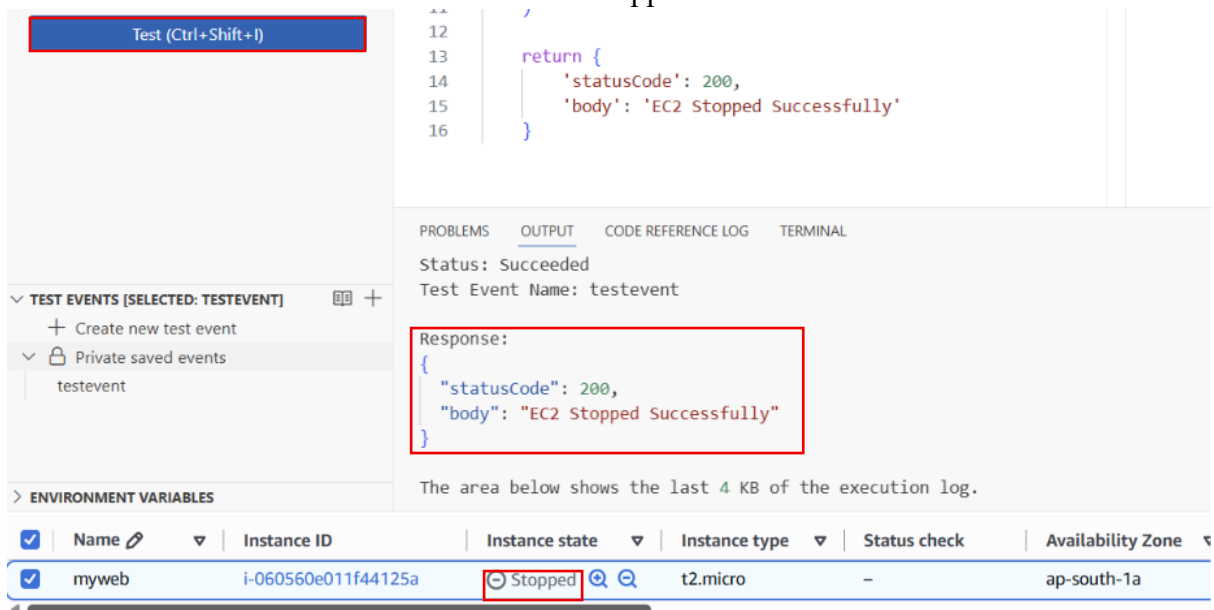
- Create another lambda function EC2 Instance to stop to the Same way
- Import source code for EC2 instance to stop
- Some changes in code, where start change into stop
- Click on the Deploy

- Create Test Event

Code source [Info](#)



- Click on the Test
- StopEC2Instance is ready to Work
- Check the EC2 Instance because instance stopped



Step 17: Add API Gateway trigger in both function

- Click on the Add trigger



- Choose API Gateway in Trigger configuration
- Create a new API
- Click on the checkbox HTTP API
- Choose Open in security

Add trigger

Trigger configuration info

☒ **API Gateway** aws api application-services backend HTTP REST serverless

Add an API to your Lambda function to create an HTTP endpoint that invokes your function. API Gateway supports REST, HTTP, and WebSocket APIs. [Learn more](#)

Intent
Use an existing API or have us create one for you.

☒ **Create a new API**
☐ Use existing API

API type

☒ **HTTP API**
Build a REST API and cost-effective REST APIs with built-in features such as OIDC and OAuth2, and native CORS support.

☐ **WebSocket API**
Build a WebSocket API using persistent connections for real-time use cases such as chat applications or dashboards.

☐ **REST API**
Develop a REST API where you gain complete control over the request and response along with API management capabilities.

Security
Configure the security mechanism for your API endpoint.

☒ **Open**

- Click on the Add

► **Additional settings**

Lambda will add the necessary permissions for Amazon API Gateway to invoke your Lambda function from this trigger. [Learn more](#) about the Lambda permissions model.

[Cancel](#) [Add](#)

- Click on the API Gateway
- Copy the API endpoint

API Gateway [+ Add destination](#) Last modified 59 minutes ago

[+ Add trigger](#)

Configuration Code Test Monitor Configuration Aliases Versions

Configuration

General configuration

Triggers

Permissions

Destinations

Function URL

Triggers (1) Info

☐ Trigger

☒ **API Gateway: StartEC2instance-API**
arn:aws:execute-api:ap-south-1:463646775279:h8xhi9qc7/*/*/*StartEC2instance
API endpoint: <https://h8xhi9qc7.execute-api.ap-south-1.amazonaws.com/default/StartEC2instance>
► Details

Step 18: Create Index.html file in desktop

- Go to Vs code open file
- Implement code from Resources
- Paste StartEC2instances API endpoint in code

```

<html lang="en">
<head>
  <style>
  </style>
</head>
<body>

  <div class="container">
    <div class="branding">CloudFolks Lambda With AI Agent</div>
    <h1>Manage Your EC2 Instance</h1>
    <button class="start" onclick="triggerAPI('start')">Start EC2</button>
    <button class="stop" onclick="triggerAPI('stop')">Stop EC2</button>
  </div>

  <script>
    function triggerAPI(action) {
      let startApiUrl = "https://h8xhi9rqc7.execute-api.ap-south-1.amazonaws.com/default/StartEC2Instance";
      let stopApiUrl = "https://p0b11huspc.execute-api.ap-south-1.amazonaws.com/default/StopEC2Instance";

      let apiUrl = action === "start" ? startApiUrl : stopApiUrl;

      fetch(apiUrl, { method: "GET" })
        .then(response => console.log("API triggered:", response))
        .catch(error => console.error("Error:", error));
    }
  </script>

</body>
</html>

```

- Add Trigger API Gateway in your StopEC2Instance function to the same way
- Copy the API endpoint of StopEC2Instance
- Paste StopEC2Instance API endpoint in code

```

<html lang="en">
<head>
  <style>
  </style>
</head>
<body>

  <div class="container">
    <div class="branding">CloudFolks Lambda With AI Agent</div>
    <h1>Manage Your EC2 Instance</h1>
    <button class="start" onclick="triggerAPI('start')">Start EC2</button>
    <button class="stop" onclick="triggerAPI('stop')">Stop EC2</button>
  </div>

  <script>
    function triggerAPI(action) {
      let startApiUrl = "https://h8xhi9rqc7.execute-api.ap-south-1.amazonaws.com/default/StartEC2Instance";
      let stopApiUrl = "https://p0b11huspc.execute-api.ap-south-1.amazonaws.com/default/StopEC2Instance";

      let apiUrl = action === "start" ? startApiUrl : stopApiUrl;

      fetch(apiUrl, { method: "GET" })
        .then(response => console.log("API triggered:", response))
        .catch(error => console.error("Error:", error));
    }
  </script>

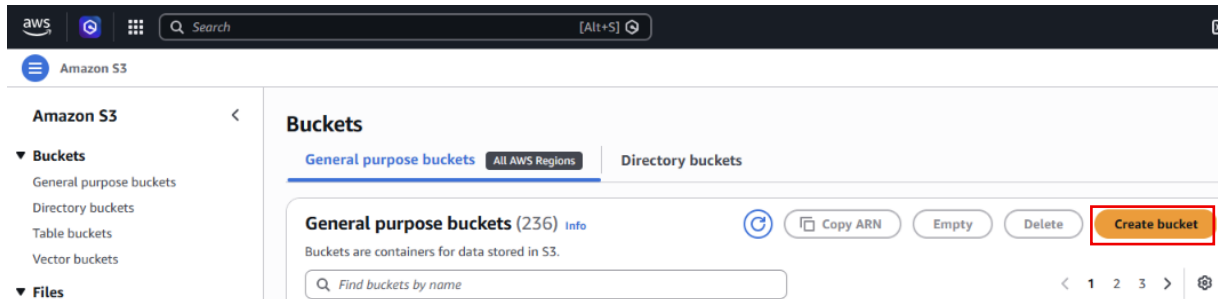
</body>
</html>

```

Step 19: Create an S3 Bucket for Web Hosting

- Go to Amazon S3 in your AWS Account

- Navigate the Bucket
- Click on the Create bucket



Step 20: Configure Bucket

- Enter the bucket name to identify the bucket
- Leave all remaining setting as default
- Click Create Button

Create bucket info

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type info

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

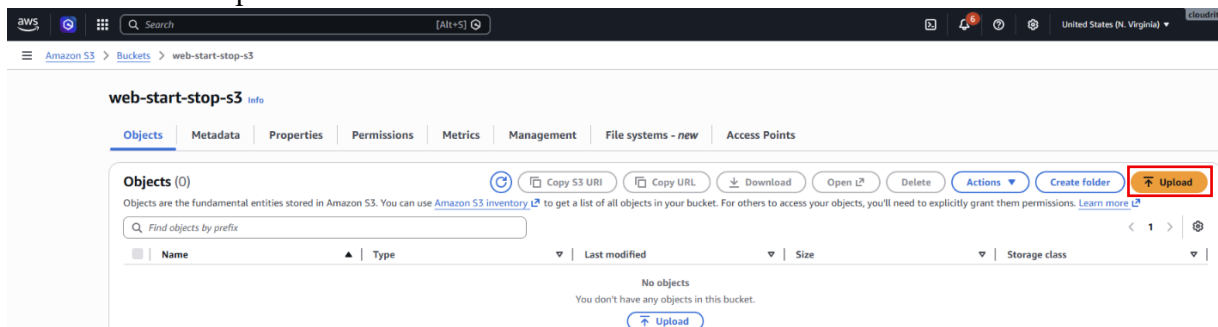
☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name info

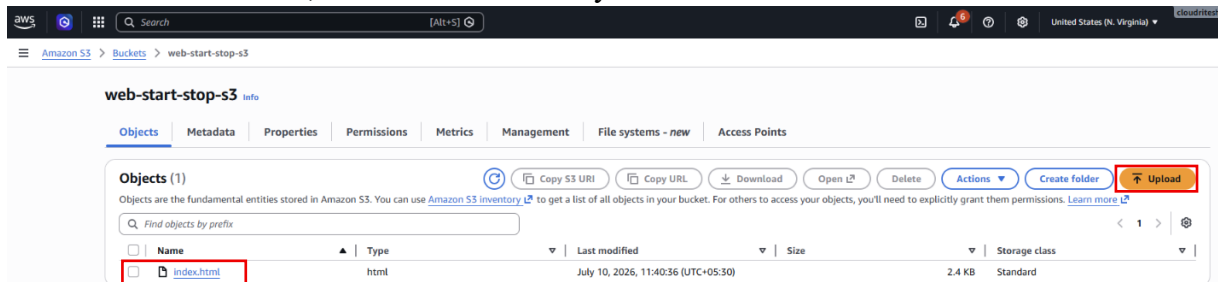
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Step 21: Upload file in bucket

- Click on the upload

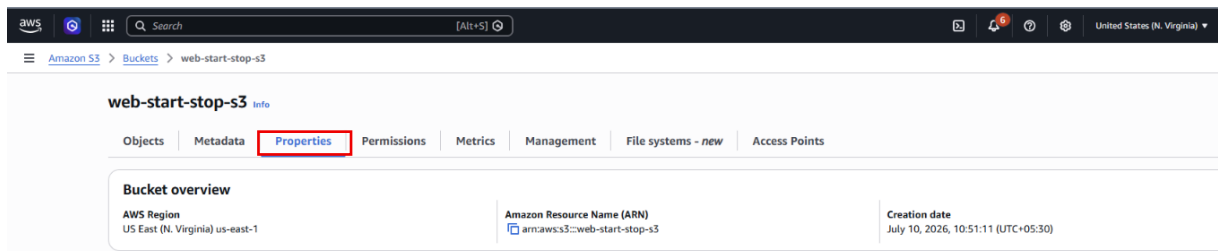


- Click on the Add file
- Add file Index.html, that who created file you before

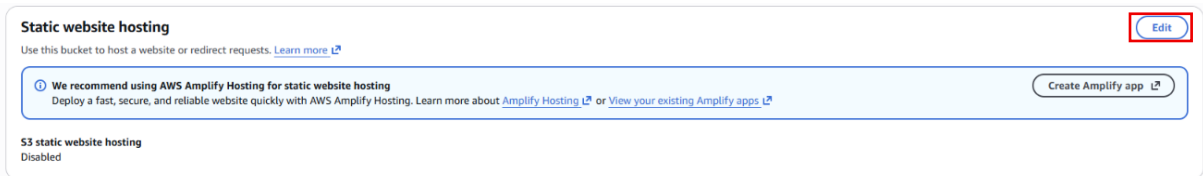


Step 22: Edit Static Website hosting

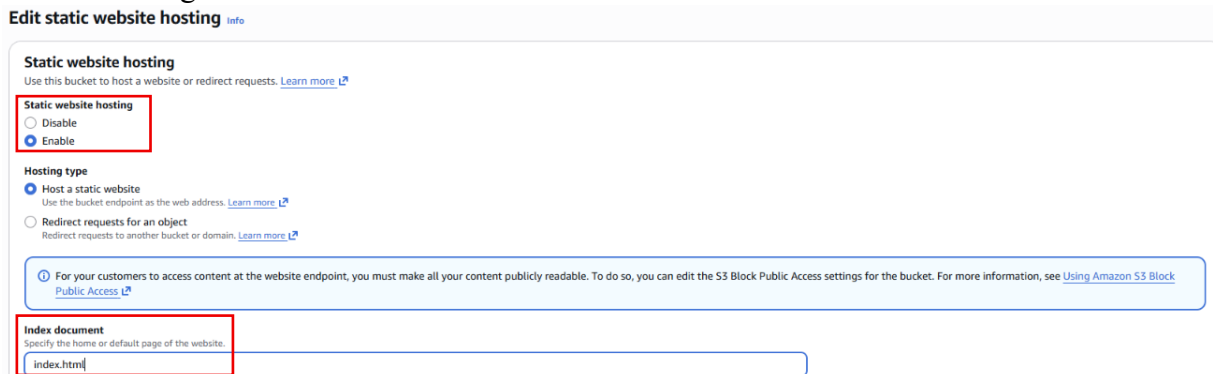
- Go to the Properties
- Copy the Amazon Resources Name (ARN)



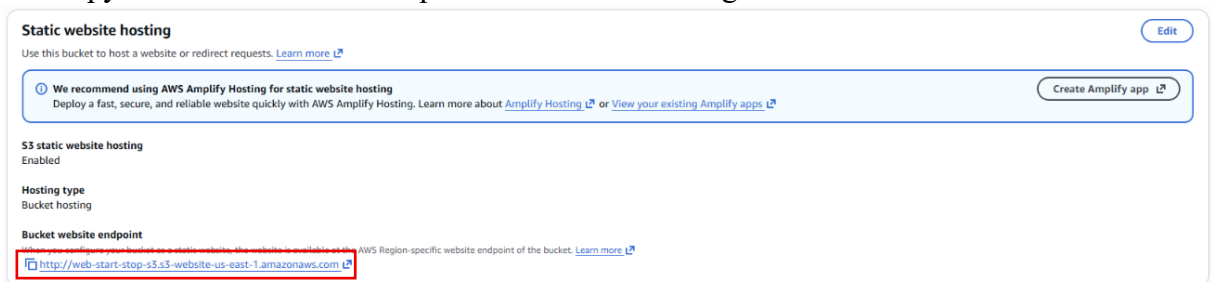
- Scroll Down
- Navigate to Static Website hosting
- Click on the Edit



- Enable the static website hosting
- Navigate to Index document, provide a file name who uploaded file
- Save changes



- Copy the bucket website endpoint url for web hosting



Step 23: Allow the permission

- Go to the own Bucket
- Navigate to permission

web-start-stop-s3 Info

Objects | Metadata | Properties | **Permissions** | Metrics | Management | File systems - new | Access Points

Permissions overview

Access finding
Access findings are provided by IAM external access analyzers. Learn more about [How IAM analyzer findings work](#) 🔗
[View analyzer for us-east-1](#)

Block public access (bucket settings) Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) 🔗

Block all public access
☒ On
 ▶ Individual Block Public Access settings for this bucket

Bucket policy Edit Delete

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#) 🔗

🔔 Public access is blocked because Block Public Access settings are turned on for this bucket
 To determine which settings are turned on, check your Block Public Access settings for this bucket. [Learn more about using Amazon S3 Block Public Access](#) 🔗

No policy to display. Copy

- Disable block all public access
- Click save changes, then go bucket policy
- Click on the edit

Edit Block public access (bucket settings) Info

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) 🔗

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Cancel **Save changes**

- Implement the code
- Paste Amazon Resources Name (ARN) in code

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadForWebsite",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "/"
    }
  ]
}
```

- Click on the Save

Edit bucket policy Info

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN
arn:aws:s3::web-start-stop-43

Policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "PublicReadGetObject",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "s3:GetObject",
9       "Resource": "arn:aws:s3:::cloudfreeks-web-demo/*"
10    }
11  ]
12 }

```

Edit statement

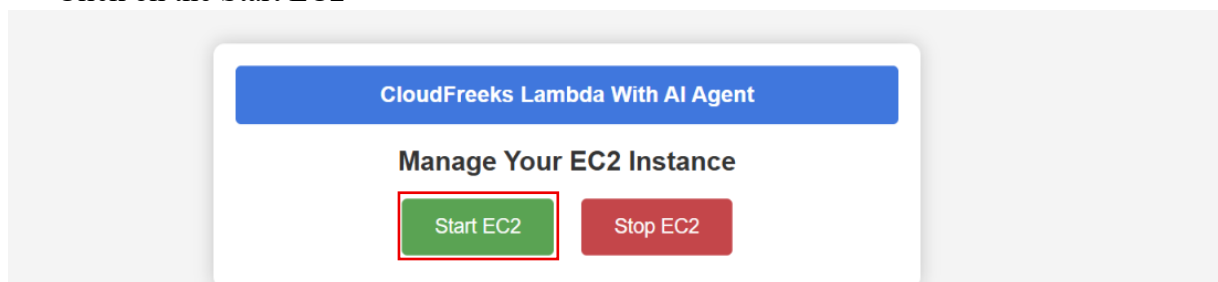
Select a statement

Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

Step 24: Verifying the Web hosting

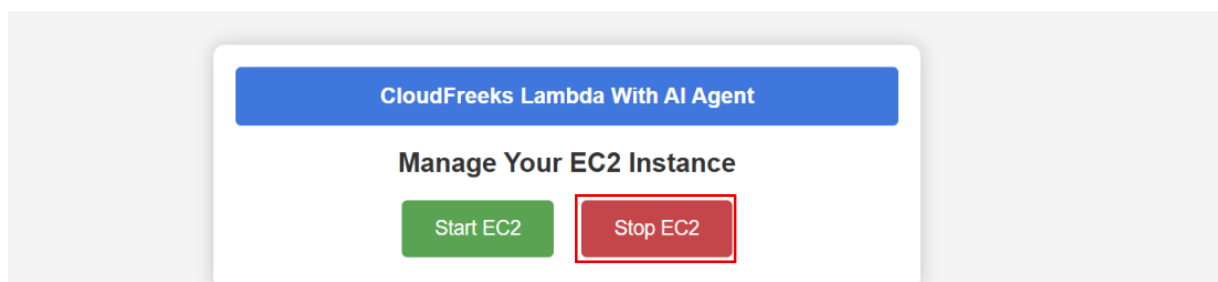
- Click on the Start EC2



- EC2 instance is Running

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	myweb	i-060560e011f44125a	Running	t2.micro	2/2 checks passed	ap-south-1a

- Click on the Stop EC2



- EC2 instance is stopped

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	myweb	i-060560e011f44125a	Stopped	t2.micro	-	ap-south-1a